Lecture 1: From a legal question to data

Empirical Methods for Law and Economics | EDLE

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Today's question

How long, and how costly, is it to enforce a commercial contract, and how much does this vary across economies?

- By the end of today you will be able to:
  - state what one row of a dataset represents and how a variable was measured;
  - describe a categorical and a quantitative variable with the right summary and its denominator;
  - read mean, median, standard deviation, quartiles and a histogram;
  - save a do-file that reproduces every number you report.

DISCUSS Why enforcement matters for Law and Economics

- Contracts are promises backed by the threat of a court. Slow or costly enforcement weakens that threat.
- It changes incentives to trade with strangers, to invest, to settle, and to breach.
- Djankov, La Porta, Lopez-de-Silanes and Shleifer (2003), "Courts", *QJE*: they measure how long the same two simple disputes (evicting a tenant, collecting a bounced cheque) take in 109 countries, and relate duration to procedural formalism and legal origin.
- Doing Business turned that method into an annual indicator. Today we use it purely to **describe**.

If enforcing a contract takes three years, what changes in how firms trade, write contracts and settle disputes?

ANSWERS Why enforcement matters

- Firms trade more with partners they know, rely on reputation and long relationships, and less with strangers.
- Contracts shift towards prepayment, collateral, guarantees and arbitration clauses.
- Creditors accept lower settlements; debtors gain from delaying or breaching.
- Djankov et al. (2003): duration is longer where procedures are more formalistic.

The data

- World Bank, *Doing Business*, "Enforcing contracts" indicators.
- Edition **Doing Business 2020**, data as of 1 May 2019. The source labels this year "2019".
- 191 economies (Liechtenstein included although outside the 190-economy ranking); regional aggregates excluded.
- Licence CC BY 4.0: values unchanged; we selected, renamed and labelled variables.
- The programme was discontinued in 2021 after an external review found data irregularities in the 2018 and 2020 editions. We treat these numbers as historical teaching data, not as a current ranking.

DISCUSS What exactly is measured: a standardized case

- A hypothetical commercial dispute over the sale of goods between two domestic firms, with a claim worth 200% of income per capita (or US$5,000 if higher).
- The case goes to the first-instance court of the **largest business city**; for 11 economies two cities are measured and averaged.
- Values come from questionnaires to local lawyers and judges and from procedural rules, not from court files.
- `time_days` = filing and service + trial and judgment + enforcement of the judgment.
- `cost_pct_claim` = court, attorney and enforcement costs as % of the claim.

What does a standardized case in the largest business city leave out?

ANSWERS What a standardized case leaves out

- Variation within the country: other cities, other courts, other kinds of dispute.
- Observed cases: values come from experts and procedural rules, not from court files.
- Cases that settle or are abandoned, and appeals.
- A claim fixed relative to income per capita: the stakes in US dollars differ across economies.

The unit of observation

- One row = one economy, in one edition.
- There are no individual court cases in this file, so we cannot describe how case durations vary *within* a country.
- The 191 economies are (almost) all economies covered, not a random sample from a larger population.
- Referee question no. 1, for any paper: *what is one row, and does the claim in the text match that level?*

DISCUSS Variables and their types

Variable	Type	Unit or scale
`economy_code`	identifier	a label: never average it
`region`	categorical (7 groups)	counts and shares
`income_group`	ordered categorical (4)	counts and shares
`time_days`	quantitative	days
`filing_days`, `trial_days`, `enforcement_days`	quantitative	days; the three sum to `time_days`
`cost_pct_claim`	quantitative	% of claim; can exceed 100
`judicial_quality`	index	points, 0–18
`court_automation`, `adr_index`	index components	points, 0–4 and 0–3

- Digits are not always quantities: a code can stand for a category. Ask what a one-unit difference means.

Which of these variables can you average, and which can you only count?

ANSWERS Average or count?

- Average: `time_days` and its three components, `cost_pct_claim`.
- Average with care: the indices (`judicial_quality` and its components), which assume every point is worth the same.
- Count and share only: `region`, `income_group` (ordered, but one step is not a unit).
- Never: `economy_code`, an identifier.

STATA Guided Stata 1: open a reproducible analysis

```
cd "C:/path/to/your/course/folder"     // your own path
do "do/00_setup.do"
use "data/contracts_2019.dta", clear
describe
codebook time_days cost_pct_claim region, compact
list economy time_days filing_days trial_days enforcement_days ///
     if economy_code=="ITA", noobs
```

- Type commands in your own do-file, then run them. The do-file is your record; the Results window is not.

STATA Guided Stata 2: categories and denominators

```
tabulate region, missing
```

Region	Economies	Percent
Europe & Central Asia	50	26.2
Sub-Saharan Africa	48	25.1
Latin America & Caribbean	33	17.3
East Asia & Pacific	29	15.2
Middle East & North Africa	21	11.0
South Asia	8	4.2
North America	2	1.0
Total	191	100.0

- "26.2% of the 191 economies in the file are in Europe and Central Asia." Always name the denominator.

PREDICT Zero, missing, and absent

How many economies score 0 on court automation? Is a 0 a missing value?

ANSWERS Zero, missing, and absent

- **Zero**: 73 economies score 0 on court automation. This is an observed score (no automation features), not missing information.
- **Missing value**: the row exists but the value is unknown (Stata shows "."). This extract has none in the selected variables.
- **Absent row**: an economy not covered is not in the file at all; `tabulate` cannot show it.
- Stata stores missing as larger than any number: `if x > 0` *includes* missing values. Write `if x > 0 & !missing(x)`.

STATA Guided Stata 3: centre and spread

```
summarize time_days cost_pct_claim, detail
```

	Mean	Median	SD	Q_1	Q_3	Max
`time_days` (days)	649.7	575	304.2	465	755	1,785
`cost_pct_claim` (%)	32.9	27.5	19.5	21.8	38.0	163.2

- Mean above median: a long right tail of slow (and costly) economies pulls the mean up.
- Timor-Leste: cost 163.2% of the claim. Enforcing costs more than the claim is worth. Unusual, not an error.

PREDICT Mean versus median: an experiment on real data

- Suppose someone mistyped Timor-Leste's cost as 1,632 instead of 163.2.

Predict the new mean and the new median before computing them.

ANSWERS Mean versus median: an experiment on real data

- Mean cost: $32.9 \;\to\; 32.9 + (1632 - 163.2)/191 = 40.6$.
- Median cost: stays at 27.5.
- The mean uses every value; the median uses only the order. Report both when they differ.
- An extreme value is a fact to explain or document, not a value to delete. Check units and source before calling it an error.

Spread: standard deviation and quartiles

- Standard deviation: typical distance from the mean, in the original units.

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

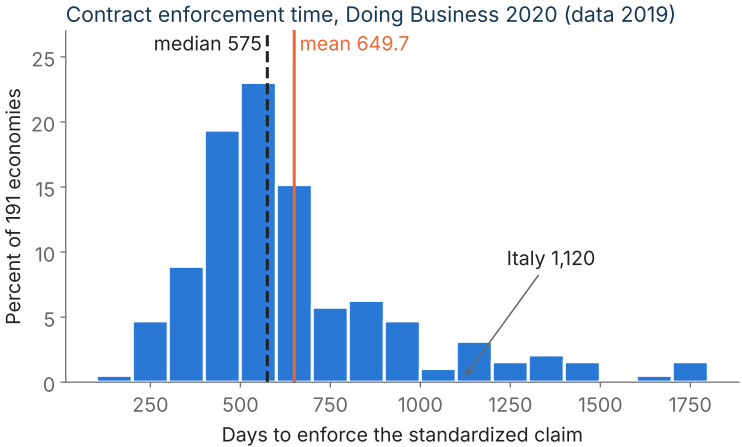
- Quartiles split the ordered values into four groups of equal size; the interquartile range $Q_3 - Q_1 = 755 - 465 = 290$ days covers the middle half.

- Here the SD (304 days) describes how different **economies** are. It says nothing about uncertainty, and nothing about variation among cases within a country. Tomorrow: the standard error.

PREDICT Shape: the histogram

Sketch the shape before you see it: where is the peak, and which tail is longer?

ANSWERS Shape: the histogram

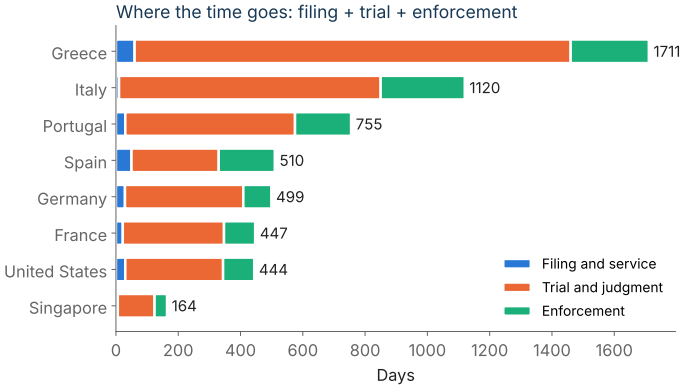


Source: World Bank Doing Business 2020 (data 1 May 2019), CC BY 4.0. 191 economies; bins of 100 days.

PREDICT Where is Italy, and where does the time go?

What share of economies is faster than Italy, and which phase dominates?

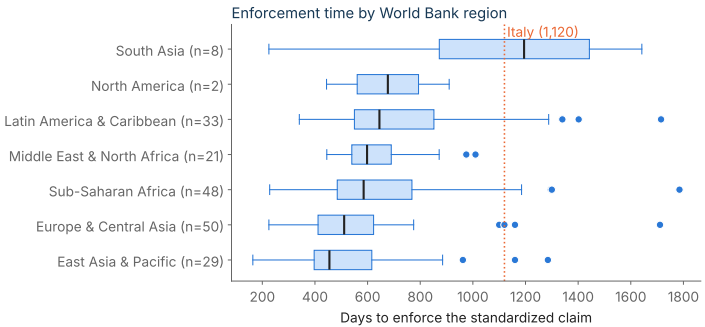
ANSWERS Where is Italy, and where does the time go?



- Italy: 1,120 days; 90% of economies take less (172 of 191). Trial and judgment: 840 days (median economy: 365).
- Cost 27.6% of the claim, close to the median: slow, not especially expensive.

STATA Comparing groups is still description

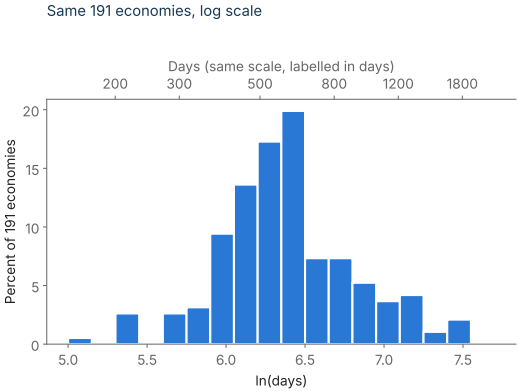
```
tabstat time_days cost_pct_claim, by(region) ///
        statistics(n mean p50) format(%9.1f)
graph hbox time_days, over(region, sort(1) descending)
```



STATA Logarithms: a change of scale

```
generate double ln_time = ///
  ln(time_days) ///
  if time_days>0 & !missing(time_days)
histogram ln_time, percent
```

- A log difference is a ratio: $\ln 1120 - \ln 575 = 0.67$, and $e^{0.67} = 1120/575 = 1.95$ (95% longer). Only for small differences is it close to the percentage change ($\ln 600 - \ln 575 = 0.043 \approx 4.3\%$).
- All 191 values are positive, so no row is lost. With zeros (e.g. awards of 0), $\ln 0$ is undefined and those rows would drop.



DISCUSS What descriptive statistics can and cannot support

For each sentence: can these data support it?

1. "In the 2020 edition, the standardized case took 1,120 days in Italy, against a median of 575 across 191 economies; most of the gap is in the trial phase."

2. "Italian civil cases last on average 1,120 days."

3. "Court automation shortens trials."

4. "This is how long it takes today."

ANSWERS What descriptive statistics can and cannot support

- **Can**: "In the 2020 edition, the standardized case took 1,120 days in Italy, against a median of 575 across 191 economies; most of the gap is in the trial phase."
- **Cannot**: "Italian civil cases last on average 1,120 days." (wrong unit and measurement)
- **Cannot**: "Court automation shortens trials." (a comparison, not a cause)
- **Cannot**: "This is how long it takes today." (historical edition)

Laboratory in pairs

Template: `do/exercises/01_data_and_description.do`. Swap typing and explaining halfway.

1. Describe the data: one row, the edition, and three variables with their units.
2. Tabulate `region` and `income_group`; write one sentence with a denominator.
3. Summarize time and cost (mean, median, SD, quartiles); explain why mean $>$ median.
4. Histogram of time and of log time; check the number of observations in each.
5. Locate Italy (or an economy of your choice) in the distribution and decompose its time.
6. Compare regions with `tabstat`; note the group sizes.
7. Write five sentences: unit and source, one share, centre and spread, one comparison, one limitation.

Exit check

1. What does one row of `contracts_2019.dta` represent?

2. Why is a court automation score of 0 not a missing value?

3. The mean time is 650 days and the median 575. What does that tell you about the distribution?

4. Why can't we say that Italian civil cases last 1,120 days on average?

For tomorrow

- Read Gneezy and Rustichini (2000), "A Fine Is a Price", *Journal of Legal Studies* 29(1): abstract, Sections I–II, Table 1 (p. 6) and Figure 1 (p. 7). About 30–45 minutes.

- Bring short notes on: what one row and one cell of Table 1 contain; what Figure 1 averages; when the fine starts and ends.

- Skip the formal tests and the theoretical model.